

Simless GSM positioning for navigation in GPS-denied environments

Inmaculada Soldado Serrano, José García Doblado, Ignazio Federico Finazzi and María Ángeles Martín Prats
Electronics Engineering Department, University of Seville, Sevilla, Spain

Abstract

Purpose – Until now, air navigation systems have mainly relied on global navigation satellite systems (GNSS) on the worldwide spread global positioning system. However, the so-called GNSS-denied environments open a new research line which pursues the development of alternative technologies which will cover this gap in positioning systems' services.

Design/methodology/approach – This paper presents the possibility of using positioning systems based on global system for mobile communications (GSM) signal. This approach developed in a standalone device will provide real-time information. The presented algorithm pursues a new methodology for providing useful information.

Findings – Among all the different technologies aimed at giving a navigation solution in the absence of any kind of GNSS in which this paper is based, it advocates for the use of the signals of opportunity, particularly the usage of GSM.

Practical implications – Technology is currently immersed in an era of continuous progress and expansion of navigation systems. These are evolving toward high performance systems, offering precise, efficient and safe air navigation. In addition, the growing demand for unmanned aerial vehicles increases the level of exigency on this activity even more. Therefore, in the context of the development of unmanned navigation technologies, the aim is to implement positioning systems that will allow high precision even in though environments.

Originality/value – Referencing the SIMless concept, a SIM-free system is described in this paper. The SIM-free system is supported by open data bases and permits the positioning based on the information sniffed from the signals broadcast by a set of several nearby base station of the GSM network. Hence, it provides same and in some cases even better accuracies than the already developed techniques, not being necessary to synchronize the link between the mobile terminal and the base station transceiver (BTS).

Keywords GPS-denied environment, GSM positioning, Navigation aid, SIMless

Paper type Research paper

1. Introduction

The recent advances in Avionic technologies are improving the demanding navigation performances which are required in the aeronautic sector nowadays. In this term, the global navigation satellite systems and a large number of alternative techniques based on satellite positioning, aided inertial navigation and beacons are under development. Furthermore, even a larger number of research lines are focusing on the development of positioning systems based on the signals of opportunity (SoOP). The main goal of these researches is providing an alternative global positioning system (GPS) support when these fundamental signals for navigation are not available, i.e. indoor spaces (Varshavsky *et al.*, 2007) or urban scenarios.

The SoOP are radio frequency (RF) signals which main purpose is not navigation. A high density of SoOP can be found in urban areas. Hence, they can be used for navigation purposes in case the GPS is not available (Raquel and Martín, 2008). Furthermore, an additional infrastructure is not necessary for their usage, as the current one can be reused.

Nevertheless, the development of a hardware and software SoOP receiver is a challenging goal which is hardly pursued by engineers. The receiver's drawbacks are given by three principal factors. The first one is related with the non-navigation purpose of the received RF signals. The second one bases this system's functionality in the uniform extension and distribution of these signals around the world. The last one makes it necessary to know where the signal transmitters are allocated, even their transmission direction in case the user receiver is far away from those BTS. Although some multipath problems are also to be taken into account, the SoOP research focused on AM radio, Wi-Fi, Bluetooth and DVT signals is providing good results in terms of positioning estimation (Mautz, 2012). Hence, they are being developed to add support to the current navigation systems. Even in a higher level in this term, it has to be mentioned the global system for mobile communications (GSM), whose research is also providing good results to the above mentioned development (Mautz, 2012). A simple algorithm for positioning based on GSM is presented in this paper.

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The GSM is a worldwide standard for the digital communication between mobiles that was developed by the European Telecommunications Standards Institute (ETSI) (ETSI, 2016). There are numerous GSM navigation techniques for locating a user terminal whose algorithms are used for positioning depending on the BTS signal range. Well-known techniques for passive emitter localization are, for example, time difference of arrival, time of arrival (Torrieri, 1984; Ho and Chan, 1993), frequency difference of arrival, angle of arrival (Schell and Gardner, 2018; Weiss, 2004) or the combination of two or more of those methods (Mušicki and Koch, 2010), among other methods (Mautz, 2012).

However, the majority of these techniques require a GSM station link and challenging algorithms. This article advocates for the use of a less difficulty algorithm based on obtaining the Cell ID which provides information about where the BTS is geographically allocated. This Cell identification is related with a radio range where the user can be situated taking into account the theoretical BTS signal strength. This coverage can vary from 100 m in urban areas up to 35 km in rural areas where the BTS density is significantly lower. In addition, the user terminal does not have to establish a communication with the base transmission station (BTS) and, therefore, it does not need to have a SIM. This concept, which is called *SIMless*, allows the user terminal to receive signal from all the near BTS no matter the phone company they belong to. Consequently, it is possible to overlap coverage areas to obtain more information for the cell identification. Thus, this article is based on a MATLAB algorithm implementation which allows the extraction of broadcast downlink GSM information for static positioning purposes.

2. Algorithm description

The pursued final GSM positioning system is able to acquire and post-process the GSM signals, offering a positioning solution. Thus, the whole development can be divided into two fundamental steps which are described in detail in the two following subsections. The first one is related to the downlink broadcast GSM signal sniffing and decoding. The second one is referred to the positioning algorithm implementation.

The extraction of the needed GSM information is done via the two first mentioned steps. Then, after using the USRP-E310 for sniffing the downlink broadcast GSM signals and after decoding these signals with the GNU Radio software, the Cell ID information is obtained from the logical BCCH message [GSM 04.03]. The BTS identification consists of two messages, the location area identity, which consists of a mobile country code, the mobile network code and de location area

code, and the Cell ID itself. They unequivocally identify each BTS after their verification with a station data base where each BTS is placed in its geographical coordinates. Then, these coordinates are given as inputs to the developed MATLAB algorithm. Finally, the algorithm implements the technique that will be presented in the second subsection providing a positioning solution.

The block diagram illustrated in Figure 1 shows the process which is followed by the GSM signals and has been described above.

2.1 Sniffing and decoding global system for mobile communications signals

These processes are supported by the USRP-E310. This device is an Ettus Research Software Define Radio which allows the RF signals processing. In the first step, the broadcast GSM signals are sniffed in the range of 921 MHz up to 960 MHz. After an ADC conversion, GNU radio allows the BCCH decoding. The simplest way to run this process is using GNU radio companion (GRC), a python code generation tool used to generate the blocks diagram which allows the decoding of the determined information. Figure 2 shows the diagram that has been used for sniffing and decoding GSM signals.

The information extracted from all the near BTS is given as inputs to the MATLAB algorithm following described.

2.2 Proposed algorithm

This subsection presents the algorithm description. The proposed algorithm uses the MATLAB toolbox image processing to obtain mask information as it is explained as following.

The flow diagram in Figure 3 is divided into four different blocks which are necessary for the correct extraction of the user positioning. Following, they will be explained.

First, the BTS LLA coordinates, i.e. latitude and longitude referred to the World Geodetic System 84 (WGS84), are obtained by validating the Cell ID information with the BTS data base provided as input. It is necessary to assign an initial radio of coverage for all the BTS whose signals are received in the user terminal. It is also pre-determined the number of final stations, i.e. the stop condition, that the algorithm calculates the position with. In this paper, that number is three. After that, the mask is built in relation with the map dimensions which are needed for locating all the near BTS.

Second, the LLA coordinates are transformed into the Universal Transverse Mercator (UTM) 2D coordinate system. Apart from this, it is necessary to transform the defined point in the UTM matrix using a lineal transformation to obtain the correct index in the built mask.

Figure 1 GSM system flow graph

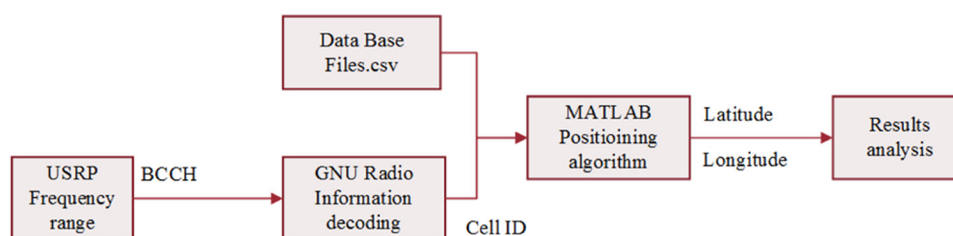


Figure 2 GRC GSM decoder

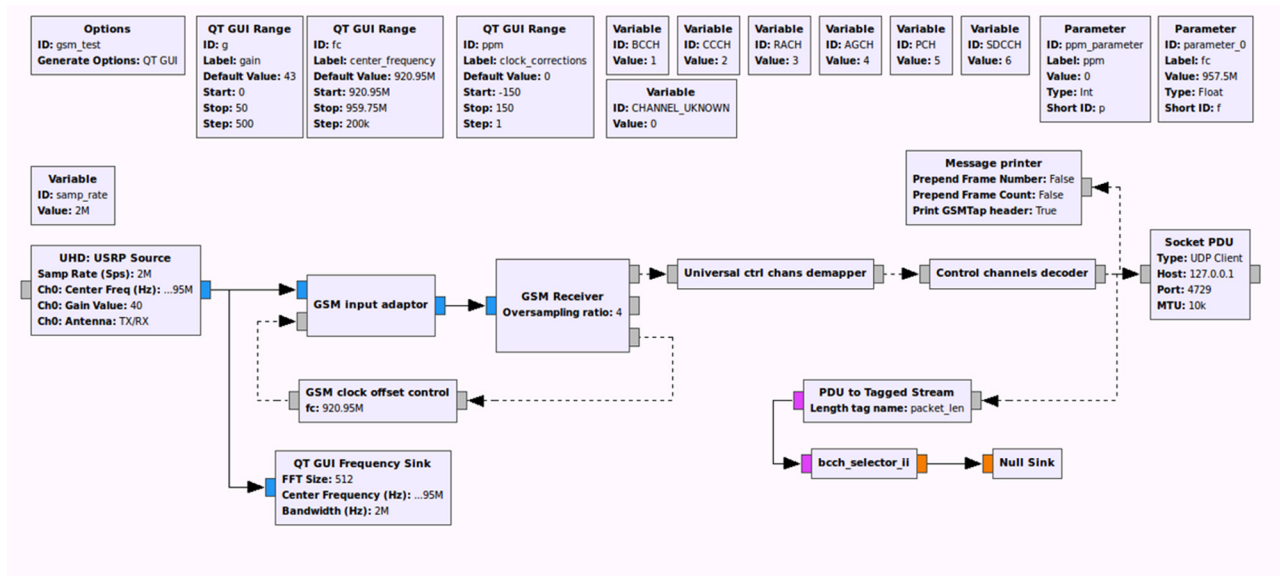
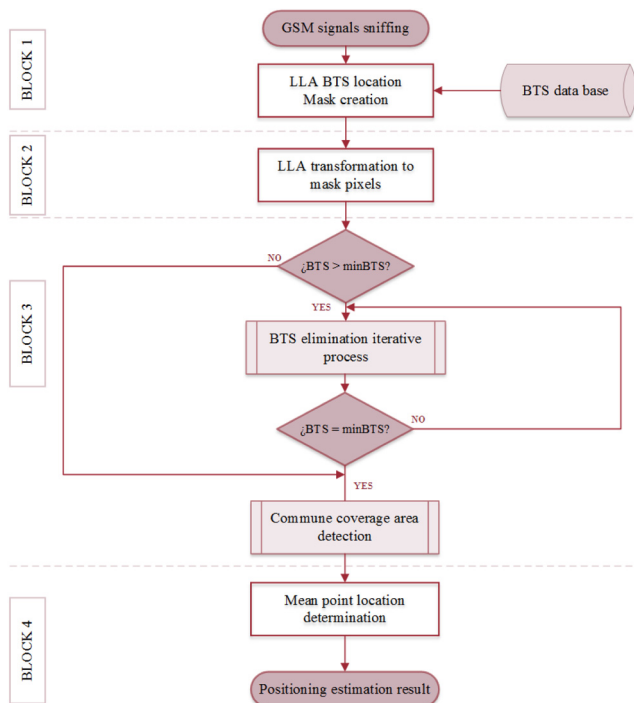


Figure 3 Proposed flow diagram for the MATLAB GSM positioning algorithm



The third block comprises the iterative process that allows for the decrease of the BTS radio coverage until a single point is determined. In addition, it takes into account the algorithm stop condition for breaking the process. The main purpose of that process is to corroborate whether after a established radio decrease it exists a non-connected BTS to the rest. In case this decision condition is true, the isolated BTS is deleted from the following iteration. If there are more than one in this situation,

the farthest BTS is first removed and the radio is reduced to continue the process. In case this decision point is false, the radio decreases again. When the minimum number of BTSs which have shared areas is reached, the mask pixels which are commune for the final BTS are calculated.

Finally, the mean point is assumed to be the positioning result, providing its transformation into LLA coordinates the positioning estimation.

3. Global system for mobile communications test example

In this subsection, an algorithm operation example is shown to simplify the above explanation. Hence, once the information is decoded and the near BTSs LLA coordinates are transformed into the mask pixels, the first scenario illustrated in Figures 4 and 5 is obtained.

Figure 4 BTS on a visual area map

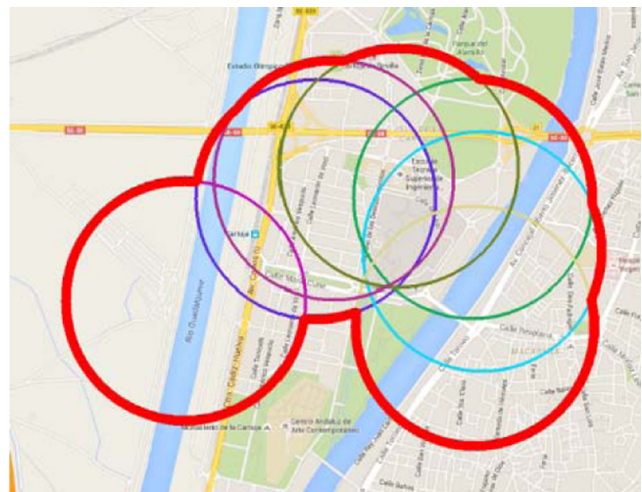
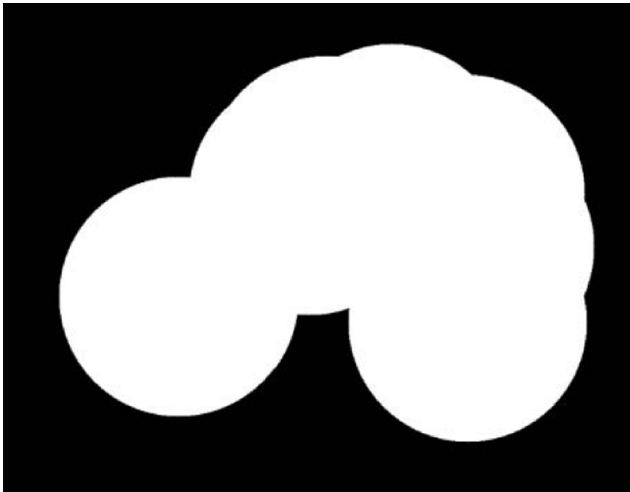
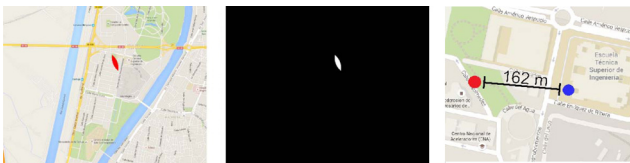


Figure 5 BTS on the determined mask

Then, this information passes through the iterative process until the coordinate point, which provides the user's positioning, is obtained as shown in [Figure 6](#). Furthermore, the meter's difference between the GPS positioning and the GSM positioning is calculated to validate the correct algorithm operation. The result gives an error of 162 m, i.e. a 162 m positioning deviation from the real location, in this specific test.

Figure 6 BTS on the determined mask

4. Global system for mobile communications positioning results

Apart from the test presented above, some others have been done to analyze the algorithm precision. [Table I](#) shows the results from the urban test where there is a high BTS density compared to the expected coordinates doing the location with a GPS receiver. In addition, the last column provides the meters of error obtained.

After that analysis, [Table II](#) provides the statistic positioning values obtained from the test. Assuming the samples' normal distribution, a mean error lower than 190 m for urban scenarios in a 95 per cent confidence interval is obtained.

5. Conclusions

The algorithm implements a low complexity functional positioning technique code that provides good values of precision for positioning in urban scenarios. In addition, the final system is expected to use downlink broadcast GSM signals which belong to a global communication standard infrastructure. Thus, it is a SIM-free (*SIMless*) system with independency from interaction with this infrastructure. Then, it is possible an indoor and GPS-denied scenarios positioning due to the GSM signals availability.

Summarizing, the pursued final GSM positioning system is able to acquire and post-process the GSM signals. This positioning solution is truly useful in GPS-denied environment, i.e. indoor or urban areas where good values of positioning estimation are obtained. Because of the fact that GPS is the most precise system in terms of positioning, the alternative positioning method based on GSM signals pursues the backup usage for guaranteeing the everywhere positioning. The less lack of available signals for the position calculating, the more precise, efficient and safe aerial navigation.

Table I Urban area test results

Urban scenario	GPS latitude	GPS longitude	BTS	GSM latitude	GSM longitude	Error[m]
Alamillo	37.412784	-5.986019	21	37.411229	-5.984559	216
Macarena	37.402865	-5.989675	13	37.403678	-5.989176	100
Santa Justa	37.392045	-5.976021	21	37.390599	-5.975123	179
Los Arcos	37.388344	-5.961449	21	37.386007	-5.962184	268
Nervión	37.385166	-5.970555	13	37.386154	-5.972537	207
Viapol	37.376925	-5.975347	17	37.377113	-5.974469	85
Bami	37.361028	-5.977228	16	37.359567	-5.976050	193
Reina Mercedes	37.361028	-5.977228	12	37.363019	-5.989219	286
San Bernardo	37.381292	-5.989165	18	37.411098	-6.003093	186
Los Remedios	37.388344	-5.961449	12	37.376930	-6.004114	179
Triana	37.383074	-6.006010	11	37.382951	-6.008171	192
Plaza Gavidia	37.395056	-5.997017	13	37.393774	-5.998421	189
ETSI	37.410891	-6.001149	12	37.360551	-5.989503	162

Table II Precision for urban scenarios in a 95% confidence interval

	Mean	Sigma	Lower end of the range [m]	Higher end of the range [m]
Urban test	187.8461538	55.0648801	157.912544	217.7797637

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Further reading

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Corresponding author

Inmaculada Soldado Serrano can be contacted at: inmsolser@gte.esi.us.es